**Part A / Multiple choice** (24 marks)

*-Be sure that you have 12 different questions*
*-Complete the table by putting (the correct answer letter) in the table*

**Q1)** Three balls are drawn randomly from 10 balls, of which 6 are red color. What is the probability that we will get two not-red colors?

a) 0.03 b) 0.7 c) 0.8 d) 0.3

**Q2)** Two independent machines A, B. Working such that p(A)=0.3, p(B)=0.6. What is the probability that at least one machine will work?

a) 0.06 b) 0.78 c) 0.12 d) 0.68

**Q3)** If $A$ and $B$ be two events in a random experiment such that $P(A)=0.3$, $P(B)=0.1$ and $p(A\cap B)=0.2$ find $P(\overline{B}\,|\,\overline{A})$

a) 7/9 b) 8/7 c)8/9 d)3/7

**Q4)** Let A be: a civil engineering student, and B: 2nd level student, in a random experiment we find p(A)=0.5, p(B)=0.5, and p(A∩B)=0.3, a student is chosen at random, find the probability that he is civil engineering such that 2nd level.

a) 0.75 b) 0.95 c) 0.07 d) 0.6

**Q5)** Let x random variable on $[0,3]$ and f (x) =1/3 the Var(x) is

a) 3/5 b) 3/7 c) 3/4 d) 5/8

**Q6)** If the probability a surgical operation will be successful is 0.8 on a certain day five operations will be performed, what is the probability that at least four will be successful?

a)0. 942 b)0.042 c)0. 142 d) 0.737

**Q7)** A student answers a multiple-choice examination of five problems by pure guessing. If there are 5 choices for each problem, what is the probability that he will get two correct answers?

a) 0.051 b) 0.2611 c) 0.7406 d) 0.205

**Q8)** If X is the number of students in the lecture per day, if the average number of students per day is 6, what is the probability that on a given day there will be exactly 4 students?

a) 0.106 b) 0.161 c) 0.134 d) 0.838

**Q9)** let X be the number of fatal accidents per day in Jordan, suppose that on the average number of fatal accidents per day is four, what is the probability that in a certain day, there will be at least four fatal accidents?

a) 0.567 b) 0.762 c) 0.314 d) 0.814

**Q10)** The grades of students are normally distributed with a mean of 38 degrees and a standard deviation 4 degrees. If a student is taken at random, what is the probability that his grade exceeds 43 degrees?

a) 0.1056 b) 0.3056 c) 0.556 d) 0.8401

**Q11)** If $X \sim N(10, 16)$, then $P(X > 20) =$

a) 0.4162 b) 0.0062 c) 0. 0228 d)0. 762

**Q12)** Given the auto-correlation function for a second-order stationary periodic process $R_{xx}(T) = 36 + \frac{4}{1+4T^2}$ the variance of the process is.

a)-1 b)6 c) 4 d)3

**Part B: Fill the Blanks** (12 marks)

**Q 13**: Type tow error is accepting $H_0$

**Q 14**: If the probability of passing an exam for each student is 0.8, in a sample of students with a mean 4, what is the probability that at most two students will pass the exam.? 0.05792

**Q 15**: If $X$ is a binomial random variable with a mean 1.6 and variance of 1.28 the $P(x \geq 2 | x \leq 6) =$ 0.49664

**Q 16:** Let $X - poisson\ (3)$ the $E(4x-1) =$ 11

**Q 17:** Let x is the number of students admitted to the master program at the university per year. If the average number of students admitted to the master's program per year is 2. What is the probability that in a given year there will be exactly 5 students. 0.0018

**Q 18**: $X \sim N(\mu, \sigma^2)$ then $P(\mu + \sigma < X < \mu + 2.05\sigma) =$ 0.15245

**Part C: Written Questions** (14 marks)

**Q 19:** Consider the distribution, with the probability density function $f_x(x) = 2e^{-2x}$, $x > 0$. Find the probability density function of y such that $Y = e^x - 3$, $y > 0$.

$f_Y(y) = 2(y+3)^{\theta-1}$, $y > 0$

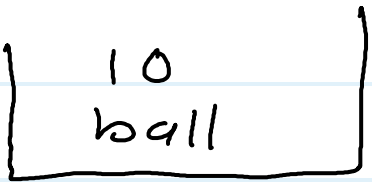
**Q 20:** Show that the process $X(t) = \cos(t + \theta)$ is first-order stationary if θ is a uniformly distributed random variable on $(0, 2\pi)$.

$E(X(t)) = 0 = \bar{X}$

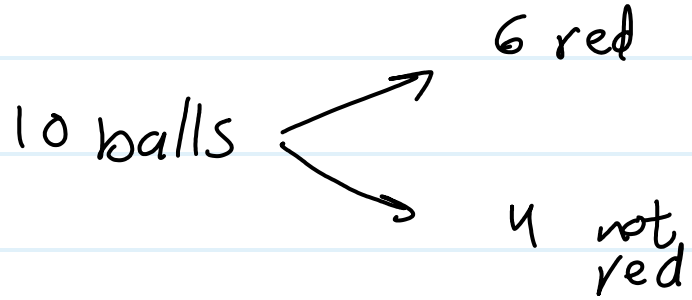
All the best /

# PART A:

Q1:




3 balls drawn $\quad \#\Omega = \binom{10}{3}$




$$p(\text{2 not red} \cap \text{1 red}) = \frac{\binom{4}{2}\binom{6}{1}}{\binom{10}{3}} = \frac{6 \cdot 6}{120}$$

$$= \frac{36}{120} = \frac{3}{10} = 0.3$$

Answer: D.

Q2: $p(A) = 0.3$ , $p(B) = 0.6$

$p(\text{at least 1 works}) = p(A \text{ works or } B \text{ works})$

$= p(A \cup B) = p(A) + p(B) - p(A \cap B)$

$= 0.6 + 0.3 - (0.3)(0.6) = 0.9 - 0.18 = 0.72$

Answer: B.

Q3: $P(\bar{B}|\bar{A}) = \frac{P(\bar{A}\cap\bar{B})}{P(\bar{A})} = \frac{1-P(A\cup B)}{1-P(A)}$

$= \frac{1-(P(A)+P(B)-P(A\cap B))}{1-0.3}$

$= \frac{1-0.2}{0.7} = \frac{0.8}{0.7} = \frac{8}{7}$ Answer: B.

Q4: $P(A) = P(B) = 0.5$

$P(A\cap B) = 0.3$ Answer: D.

such that = condition

$P(A|B) = \frac{P(A\cap B)}{P(B)} = \frac{0.3}{0.5} = \frac{3}{5}\times\frac{2}{2} = 0.6$

Q5: $Var(X) = EX^2 - (EX)^2$

$= \int_0^3 \frac{x^2}{3}\,dx - \left(\int_0^3 \frac{x}{3}\,dx\right)^2$

$= \frac{x^3}{9}\Big]_0^3 - \left(\frac{x^2}{6}\Big]_0^3\right)^2 = \frac{27}{9} - \left(\frac{3}{2}\right)^2$

$= 0.75 = \frac{3}{4}$ Answer: C.

Q6: $X \sim B(5, 0.8)$, $q = 0.2$.

$p(\text{at least } 4) = p(X=4) + p(X=5)$

$= 5\binom{5}{4} 0.8^4 0.2 + \binom{5}{5} 0.8^5$

$= 0.8^4 (5 \times 0.2 + 0.8) = 0.8^4 \times 1.8 = 0.737$

Answer: D.

Q7: Multiple choice EXAM:-

| | A. | B. | C. | D. | E. |
|---|---|---|---|---|---|
| Q1: | | | | | |
| ⋮ | | | | | |
| Q5: | | | | | |

each question is an experiment

$p(\text{right answer}) = \frac{1}{5} \times \frac{2}{2} = 0.2$

with fixed success & fail probabilities.

the experiment is repeated 5 times.

$X \sim B(5, 0.2)$

$$p(X=2) = \binom{5}{2}\, 0.2^2\, 0.8^3 = 0.2048 \approx 0.205$$

Answer: D.

Q8: students in lecture per day ⟶ poission

avg = m = 6 ⟶ $X \sim$ poission (6)

$$p(X=4) = \frac{e^{-6}\, 6^4}{4!} = 0.1338 \approx 0.134$$

Answer: C.

Q9: $Y \sim$ Poission (4)

$p$(at least 4) = $p$(4 or more) = $p(Y \geq 4)$

$= 1 - p(X<4) = 1 - (p(X=3) + \cdots + p(X=0))$

$$= 1 - \left( \frac{e^{-4}\, 4^3}{3!} + \frac{e^{-4}\, 4^2}{2!} + \frac{e^{-4}\, 4}{1!} + \frac{e^{-4}}{0!} \right)$$

$$= 1 - e^{-4}\left( \frac{32}{3} + 8 + 4 + 1 \right) = 1 - \frac{71}{3} e^{-4}$$

$= 0.5665 \approx 0.567$ Answer: A.

Q10: X : grades of students.

mean = $\mu = 38$ , std X = $\sigma_X = 4$.

exceeds ⟶ يزيد عن

$P(X > 43) = 1 - P(X \leq 43)$ , $z = \frac{43-38}{4} = 1.25$

$= 1 - P(z \leq 1.25) = 1 - 0.89435 = 0.10565$

~ using z- table ~ Answer: A.

Q11: $P(X > 20) = 1 - P(X \leq 20)$

$z = \frac{20-10}{4} = \frac{10}{4} = 2.5$

$= 1 - P(z \leq 2.5) = 1 - 0.99379 = 0.00621$

Answer: B

Q12: $C_{XX}(\tau) = R_{XX}(\tau = 0) - R_{XX}(\tau \longrightarrow \infty)$

$= 36 + 4 - 36 = 4$

Answer: C.

# PART B:

Q13: accepting $H_0$.

Q14: $p = 0.8$, $q = 0.2$.

$$EX = \mu = np \longrightarrow n = \frac{\mu}{0.8} = 5$$

$$p(2 \text{ at most}) = p(X=2) + p(X=1) + p(X=0)$$

$$= \binom{5}{2} 0.8^2 0.2^3 + \binom{5}{1} 0.8\, 0.2^4 + \binom{5}{0} 0.8^0 0.2^5$$

$$= 0.2^3 \left( 10\, 0.8^2 + 5\, 0.8\, 0.2 + 0.2^2 \right)$$

$$= 0.008\ (6.4 + 0.8 + 0.04) = 0.008 \times 7.24$$

$$= 0.05792 \quad \#$$

Q15: $\dfrac{Var(X)}{EX} = \dfrac{qpn}{pn} = q = \dfrac{1.28}{1.6} = 0.8$

So, $p = 0.2$. $\quad n = \dfrac{EX}{p} = \dfrac{1.6}{0.2} = 8$

$$P(X \geq 2 \mid X \leq 6) = \frac{P(X \geq 2 \cap X \leq 6)}{P(X \leq 6)}$$

$$= \frac{P(2 \leq X \leq 6)}{P(X \leq 6)} = \frac{0.49659904}{0.99991552} = 0.496641$$

$$\rightsquigarrow P(X \leq 6) = 1 - P(X > 6) = 1 - (P(X=7) + P(X=8))$$

$$= 1 - \left( \binom{8}{7} 0.2^7 \, 0.8 + \binom{8}{8} 0.2^8 \, 0.8^0 \right)$$

$$= 1 - 0.2^7 (8 \times 0.8 + 0.2) = 1 - 0.2^7 \times 6.6$$

$$= 0.99991552$$

$$\rightsquigarrow P(2 \leq X \leq 6) = P(X \leq 6) - (P(X=1) + P(X=0))$$

$$= P(X \leq 6) - \left( \binom{8}{1} 0.2 \, 0.8^7 + \binom{8}{0} 0.2^0 \, 0.8^8 \right)$$

$$= P(X \leq 6) - 0.8^7 (8 \times 0.2 + 0.8)$$

$$= P(X \leq 6) - 0.50331648 = 0.49659904$$

Q16: $4EX - 1 = 4m - 1 = 12 - 1 = 11$

Q17: $X \sim Poission(2)$

$$P(X=5) = \frac{e^{-2}\,2^5}{5\times4\times3\times2} = \frac{2^2\,e^{-5}}{5\times3} = \frac{4e^{-5}}{15} \approx 0.0018$$

Q18: $P(\mu+\sigma < X < \mu+2.5\sigma)$

$$z_1 = \frac{\mu+\sigma-\mu}{\sigma} = 1 \qquad z_2 = \frac{\mu+2.5\sigma-\mu}{\sigma} = 2.5$$

$$P(1 < Z < 2.5) = P(X<2.5) - P(X<1)$$

~using z-table~ $= 0.99379 - 0.84134$

$= 0.15245$ #

PART C:

Q19: $f_X(x) = 2e^{\theta x}$, $Y = e^x - 3$

$$x = \ln(Y+3) \longrightarrow \frac{dx}{dy} = \frac{1}{Y+3}$$

$$f_Y(Y) = f_X(T^{-1}(Y)) \left|\frac{dx}{dy}\right|$$

$$f_Y(y) = 2e^{\ln(y+3)^\theta} \left| \frac{1}{y+3} \right| \leftarrow y>0$$ always positive

$$f_Y(y) = 2\frac{(y+3)^\theta}{y+3} = 2(y+3)^{\theta-1}, \quad y>0 \quad \#$$

Q20: $\theta \sim U(0, 2\pi)$, $f(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \le \theta \le 2\pi \\ 0, & o.w \end{cases}$

$$X(t) = \cos(t+\theta)$$

- 1st order-stationary: $E(X(t)) = \bar{X} \equiv$ c.v

$$E(X(t)) = \int_{-\infty}^{\infty} X(t) f(\theta) d\theta = \int_0^{2\pi} \frac{A}{2\pi} \cos(t+\theta) \, d\theta$$

$$= \frac{A}{2\pi} \sin(t+\theta) \Big|_0^{2\pi} = \frac{A}{2\pi} \left( \sin(t+2\pi) - \sin t \right)$$

- $\sin(a+b) = \sin a \cos b + \cos a \sin b$

$\sin(t+2\pi) = \sin t \cos 2\pi + \cos t \sin 2\pi$ (0)

$\sin(t+2\pi) = -\sin t$ (−1)

$$\Rightarrow \frac{A}{2\pi} (-\sin t + \sin t) = 0 = \bar{X} \quad \#$$